

Summary

This video elucidates foundational concepts in probability theory, focusing on **Conditional Probability**, the **Law of Total Probability**, and **Bayes' Theorem**.

Key insights:

- Transition from basic probability to **dependent events**, where past outcomes influence future probabilities
- Use of **work-flow/tree diagrams** for path-based probability calculations
- "Favorable path over total path" approach for inverse probability problems

1. Conditional Probability

Conditional probability calculates the likelihood of an event occurring **given that another event has already occurred**.

1.1 Fundamental Logic

- **Definition:** If E1 and E2 are events, $P(E2 \mid E1)$ is the probability of E2 happening given E1 has occurred
- **Data Science Connection:** Conditional probability = Joint Probability ÷ Marginal Probability
- **Formula:**
$$P(E2 \mid E1) = \frac{P(E1 \cap E2)}{P(E1)}, \quad P(E1) \neq 0$$

1.2 Scenario: Sampling Without Replacement

Event	Scenario Detail	Probability Calculation
First Draw (Y1)	Drawing a yellow ball from 2 yellow and 3 red balls	2/5
Second Draw (Y2)	Drawing a second yellow ball after the first is removed	1/4
Combined Event	Both Y1 and Y2 occurring	$P(Y1) \times P(Y2)$

2. The Law of Total Probability

The law calculates the probability of an event **via multiple mutually exclusive paths**.

2.1 Theoretical Framework

If event A is associated with exhaustive and mutually exclusive events $E_1, E_2, \dots, E_n$, then:

$$P(A) = \sum_{i=1}^n P(E_i) \cdot P(A \mid E_i)$$

2.2 Work-Flow Diagram Methodology

1. Identify the **initial partition** (e.g., Bag 1 vs. Bag 2)
2. Assign probabilities to each branch
3. Determine the conditional probability of target event A along each branch
4. Multiply probabilities along each path
5. Sum all paths to find total $P(A)$
- 6.

3. Bayes' Theorem (Inverse Probability)

Bayes' Theorem calculates **the probability of the cause given the effect**.

3.1 Core Concept

- Total Probability asks: *"What is the chance that A occurs?"*
- Bayes asks: *"Given that A has occurred, what is the probability it came from E_i ?"*

Formula:

$$P(E_i \mid A) = \frac{P(E_i) \cdot P(A \mid E_i)}{P(A)}$$

3.2 Simplified Problem-Solving Rule

$$\text{Result} = \frac{\text{Favorable Path}}{\text{Total Paths}}$$

4. Application Examples and Case Studies

4.1 Ticket Digit Analysis

- Ticket numbers: 00–99
- Event A: Sum of digits = 7
- Event B: Product of digits = 0

- Outcomes:
 - Total Product 0 cases = 19
 - Intersection ($\text{Sum}=7 \cap \text{Product}=0$) = {07, 70}
- Calculation:

$$[P(\text{Sum}=7) \mid \text{Product}=0] = 2/19$$

4.2 Multi-Box Ball Extraction

- Three boxes with White (W) and Red (R) balls
- Goal: Probability that sequence White→Red came from Box 2
- Workflow:
 1. Assign 1/3 probability to selecting each box (B1, B2, B3)
 2. Calculate $P(W \cap R)$ for each box without replacement
 3. Apply Bayes' Theorem: isolate B2 path ÷ sum of all paths

4.3 Examination Behavior Model

- Determine probability student **knew** the answer given it was answered correctly

Methods of answering:

Method	Probability
Guessing	1/2
Copying	1/6
Knowing	$1 - (1/2 + 1/6) = 1/3$

- Probability of correctness given method:

Method	$P(\text{Correct} \mid \text{Method})$
Guessing	1/4
Copying	1/8
Knowing	1

- Apply "Favorable Path / Total Path":

$$[P(\text{Knowing} \mid \text{Correct})] = \frac{\text{Knowing path}}{\text{Sum of Knowing, Guessing, Copying paths}}$$

5. Critical Insights for Engineering Branches

- **Machine Learning:** Bayes' Theorem is the foundation of the Naïve Bayes classifier
- **System Dependence:** Understanding dependent events is crucial for real-world modeling
- **Visualization:** Use structural diagrams to account for all paths before applying inverse probability

